

Core CPU Operations

1 Instruction Cycle: Core CPU Operations

1. Fetch

- The Control Unit (CU) fetches the instruction from memory (RAM or cache).
- The Program Counter (PC) holds the address of the next instruction to be fetched.

2. Decode

- The Instruction Decoder interprets the fetched instruction.
- It determines the operation to be performed and the operands required.

3. Execute

- The ALU (Arithmetic Logic Unit) or FPU (Floating Point Unit) performs the operation (e.g., add, subtract, AND, OR).
- This may involve accessing registers or memory.

4. Memory Access (if needed)

- If the instruction requires data fetch or storage, the memory is accessed accordingly.

5. Write-back

- The result of the operation is written back to a register or memory location.

2 CPU Components Involved

Component	Role
Program Counter (PC)	Holds the address of the next instruction to execute.
Instruction Register (IR)	Holds the currently fetched instruction.
Control Unit (CU)	Directs operations of the CPU components.
ALU	Performs arithmetic and logical operations.
Registers	Fast, local data storage within the CPU.
Bus Interface	Connects the CPU to memory and I/O subsystems.

Table 1: Key CPU Components and Their Roles

3 Instruction Cycle: Continuous Repetition

Instruction Cycle Loop

The CPU continually repeats the **fetch-decode-execute** cycle at high speed, often millions or billions of times per second. This loop is driven by the system clock and ensures the continuous execution of program instructions.

4 Example Instruction: ADD R1, R2, R3

- **Fetch:** The instruction ADD R1, R2, R3 is fetched from memory.
- **Decode:** The CPU determines it needs to add contents of registers R2 and R3.
- **Execute:** The ALU performs the addition operation: $R1 \leftarrow R2 + R3$.
- **Write-back:** The result is stored in register R1.

Tip: The ADD instruction does not require memory access unless operands are stored in memory instead of registers.